

Program : Diploma in Communication and Computer Networking	
Course Code : 6321A	Course Title: Android programming
Semester : 4	Credits: 4
Course Category: Engineering Science	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Understand the Android platform and its components
- Learn the fundamentals of Java programming language
- Learn how to design and develop mobile applications using Android Studio
- Learn how to publish apps on the Google Play Store

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge in Computer systems	1008	Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

CO	Description	Duration (Hours)	Cognitive Level
CO1	Acquire knowledge of different android platforms	14	Understanding
CO2	Learn the basic android components	14	Applying
CO3	Develop knowledge about user interfaces	14	Applying
CO4	Learn how to design and develop mobile application.	16	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1				
CO2	3	1	2				
CO3	3	2	1				
CO4	3	2	1				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Acquire knowledge of different android platforms		
M1.01	To describe various mobile technologies	4	Understand
M1.02	To explain android virtual device	3	Understand
M1.03	To describe awareness about IDE	3	Understand
M1.04	To describe features of Android studio	4	Apply
Contents: Basic Computer Architecture Various mobile technologies- Apple IOS – Android operating system- install and configure Eclipse, Android Studio and Android sdk - android virtual device- creation of android virtual device- sample programs – features of Eclipse and Android studio.			
CO2	Learn the basic android components		
M2.01	To describe activities and intents	7	Understanding
M2.02	To describe android virtual device	7	Understanding
	Series Test –I	1	

Contents: Sample programs- Operation of Android Virtual device - activity in android –Life cycle of an activity intent – linking activities using intent- data passing between activities using intent - android components: activities, services, broadcast receivers, content providers. -			
CO3	Develop		
M3.01	To describe different layouts.	4	Understanding
M3.02	To illustrate different buttons and patterns	5	Understanding
M3.03	To explain different menus	5	Applying
Contents: UI Design and Data storageUI components: -Layout: Linear, Absolute, Table, Frame. - Views: Text, Edit, Button, ImageButton, CheckBox, ToggleButton, RadioButton, RadioGroup, List, Image, Grid . Menus – Options, Context- Action bar, Notifications			
CO4	Implement		
M 4.01	To describe datastorage	8	Understanding
M 4.02	To deploy an application	8	Applying
	Series Test – II	1	
Contents: data storage in Android- various storage technologies- operations for data storage and retrieval to/from internal and external memory - SQLite database- content Providers and their relative advantages and disadvantages - SMS service in Android - publish application in Google Play Store.			

Text / Reference

T/R	Book Title/Author
T1	1. Beginning Android Application Development- Wei-Meng Lee- Wrox-First Edition
T2	Android Development on Android studio-Eloquent Droid-Third Edition
T3	Android Programming for bgrinners-JohnHorton-3 rd Edition

R1	Fundamentals of Android app development-Sujith Kumar Mishra
R2	Android Programming for beginners-Tim Warren

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com/android/index.html
2	https://www.javatpoint.com/android-tutorial
3	https://developer.android.com/training/basics/firstapp
4	https://www.javatpoint.com/android-tutorial